PRECALCULUS · UNIT 4: TRIGONOMETRIC FUNCTIONS

Daylight and Seasons

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students model daylight and temperature in Boston. Earth's orbit motivates radians and the unit circle, data cards give amplitude, midline, period, and phase shift, and trig equations find event dates.

Comparing daylight with temperature shows why phase shift matters.

Standards Addressed

- F-TF.A.1–2 Radians and the unit circle
- F-TF.B.5 Model periodic phenomena with trig functions
- F-TF.B.7 Solve trig equations using inverses
- F-TF.C.8 Pythagorean identity

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with sunrise and sunset times. Part 1 unit circle.
Day 2	Part 2 building models.
Day 3	Part 3 solving for dates and the report.

Materials

- Student handout and Data Cards
- Graphing calculators or Desmos
- Optional: sunrise/sunset data from timeanddate.com

How the Three Versions Differ

Level	What changes
CP2	Unit-circle values, midline, amplitude, period, evaluating a given model, and a table search.
CP1	Exact values, building cosine models, phase shift meaning, and solving trig equations.
Honors	Equivalent models with identities, comparing cities, trig inequalities with inverses, and combined windows.

Watch For (Common Misconceptions)

- Using degrees in a radian model.
- Confusing period with frequency ($\frac{2\pi}{B}$).
- Finding only one solution of a trig equation.

Extension Ideas

- Collect real daylight data for your town.
- Model tides or a Ferris wheel.
- Compare cities in different hemispheres.

Answer Key – CP2

Part 1: Earth's Orbit

#	Task	Answer
1	How many degrees does Earth travel in one day? (Round to two decimals.)	$\frac{360}{365} \approx 0.99°$
2	Convert 90° and 180° to radians.	$\frac{\pi}{2}$ and π
3	Find $\sin\frac{\pi}{6}$, $\cos\frac{\pi}{3}$, and $\sin\frac{\pi}{2}$ using the unit circle.	$\frac{1}{2}, \frac{1}{2}, 1$
4	In which quadrant is an angle of $\frac{5\pi}{4}$? Give its sine and cosine signs.	Quadrant III; both negative

Part 2: Daylight Model

#	Task	Answer
5	Find the midline (average of max and min daylight).	$\frac{15.3+9.1}{2} = 12.2$ h
6	Find the amplitude.	$\frac{15.3-9.1}{2} = 3.1$ h
7	What is the period in days?	365
8	A model is $D(t) = 3.1 \sin(\frac{2\pi}{365}(t - 80)) + 12.2$. Find D(80) and D(172).	D(80) = 12.2 h; D(172) ≈ 12.2 + 3.1 sin(1.58) ≈ 15.3 h

Part 3: Plan Events

#	Task	Answer
9	Find D(1) (January 1) and D(200) (mid-July).	D(1) ≈ 9.2 h; D(200) ≈ 14.9 h
10	Use a table to estimate the first day with at least 13 hours of daylight.	About day 96 (early April)
11	Recommend three event dates and explain using the model.	Answers vary; dates between about day 96 and day 247 (early April to early September).

Reflection

#	Prompt	Look for
R1	Why is a sine function a good model for daylight?	Daylight repeats every year and rises and falls smoothly.

#	Prompt	Look for
R2	What does the midline represent?	The average daylight over the year (about 12.2 hours).

Answer Key – CP1

Part 1: Unit Circle

#	Task	Answer
1	Through what angle in radians does Earth travel in 73 days?	$\frac{73}{365} \times 2\pi = \frac{2\pi}{5} \approx 1.26$
2	Find the exact values of $\sin\frac{2\pi}{3}$, $\cos\frac{7\pi}{6}$, and $\tan\frac{5\pi}{4}$.	$\frac{\sqrt{3}}{2}$, $-\frac{\sqrt{3}}{2}$, 1
3	A point on Earth's orbit (radius 1 AU) is at angle $\frac{5\pi}{3}$. Find its coordinates.	$(\frac{1}{2}, -\frac{\sqrt{3}}{2})$
4	Explain why $\sin\theta = \sin(\pi - \theta)$ using the unit circle.	The points are reflections across the y-axis and have the same y-coordinate.

Part 2: Build the Models

#	Task	Answer
5	Write a cosine model for daylight with its maximum on day 172.	$D(t) = 3.1\cos(\frac{2\pi}{365}(t - 172)) + 12.2$
6	Write a cosine model for average high temperature (max 74°F on day 205, min 29°F).	$T(t) = 22.5\cos(\frac{2\pi}{365}(t - 205)) + 51.5$
7	Find D(100) and T(100).	D(100) ≈ 12.2 + 3.1 cos(−1.239) ≈ 13.2 h; T(100) ≈ 51.5 + 22.5 cos(−1.817) ≈ 46.2°F
8	Why do the daylight and temperature models have different phase shifts?	Temperatures lag behind daylight because land and water take time to heat up.

Part 3: Solve for Dates

#	Task	Answer
9	Solve D(t) = 13 for the two days in the year.	$\cos(\ldots) = \frac{0.8}{3.1} \approx 0.258 \rightarrow t - 172 \approx \pm 76.3 \rightarrow t \approx 96$ and 248
10	Solve T(t) = 60 for the two days.	$\cos(\ldots) = \frac{8.5}{22.5} \approx 0.378 \rightarrow t - 205 \approx \pm 68.7 \rightarrow t \approx 136$ and 274
11	Find the event window where both conditions hold, and give it as calendar dates.	Days 136 to 248: about May 16 to September 5

Reflection

#	Prompt	Look for
R1	How did you find the phase shift from the data?	Shift the cosine so its maximum occurs on the day of the maximum.
R2	Why does a trig equation have two solutions in a year?	The function crosses a value once going up and once going down each period.

Answer Key – Honors

Part 1: Model & Compare

#	Task	Answer
1	Write sine and cosine models for daylight with the same graph. Show they are equal with an identity.	$D = 3.1\cos\left(\frac{2\pi}{365}(t - 172)\right) + 12.2 = 3.1\sin\left(\frac{2\pi}{365}(t - 80.75)\right) + 12.2$, using $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ (91.25 days)
2	Miami has max 13.7 h and min 10.5 h. Write its model (same phase) and compare amplitudes. What causes the difference?	$M(t) = 1.6\cos\left(\frac{2\pi}{365}(t - 172)\right) + 12.1$; smaller amplitude because Miami is closer to the equator.
3	On what days do Boston and Miami have the same daylight?	Where $3.1c + 12.2 = 1.6c + 12.1 \rightarrow c = -\frac{0.1}{1.5} \approx -0.067 \rightarrow t - 172 \approx \pm 95.2 \rightarrow t \approx 77$ and 267 (near the equinoxes)
4	Find the rate of change of Boston daylight in minutes per day near the equinox (use the average rate from day 75 to 85).	$(D(85) - D(75)) \div 10 \approx 0.053$ h ≈ 3.2 minutes per day

Part 2: Inverse Trig

#	Task	Answer
5	Solve $D(t) \geq 14$ for one year.	$\cos(\ldots) \geq \frac{1.8}{3.1} \rightarrow |t - 172| \leq 55.3 \rightarrow$ days 117 to 227
6	Explain why $\cos^{-1}$ gives only one angle and how you found the second.	$\cos^{-1}$ returns an angle in $[0, \pi]$; the second solution is the negative (reflected) angle within the period.
7	Write the daylight model as a function of the angle of Earth's orbit θ (radians) instead of days.	$D(\theta) = 3.1\cos(\theta - \theta_0) + 12.2$ with $\theta_0 = \frac{2\pi(172)}{365}$
8	Prove $\sin^2 x + \cos^2 x = 1$ with the unit circle and use it to find sin θ if cos θ = 0.6 and θ is in Quadrant IV.	$x^2 + y^2 = 1$; sin θ = −0.8

Part 3: Recommendation

#	Task	Answer
9	Find the days with daylight ≥ 13 h AND temperature between 60°F and 72°F.	Daylight ≥ 13: days 96–248; T ≥ 60: days 136–274; T ≤ 72: $|t - 205| \geq 24.7 \rightarrow$ up to day 180 or from day 230. Windows: days 136–180 and 230–248.
10	Convert your windows to calendar dates.	About May 16 – June 29 and August 18 – September 5
11	Write the seasonal planning report with graphs of both models and your recommendation.	Answers vary.

Reflection

#	Prompt	Look for
R1	What does the amplitude of the daylight model tell you about latitude?	Places farther from the equator have larger swings in daylight.
R2	Why are trig inequalities solved by finding equality points first?	The function changes from above to below the value only at those points, giving the interval endpoints.